IEEE Software Project Management Plan (SPMP)

Cloud Migration of a Mid-Sized Retail Company's Legacy On-Premise ERP System to Microsoft Azure

(including Power BI Analytics Layer and Security Hardening)

Title Page

Project Title

Cloud Migration of a Mid-Sized Retail Company's Legacy On-Premise ERP System to Microsoft Azure (including Power BI Analytics Layer and Security Hardening)

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Prepared by

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Organisation

Cape Peninsula University of Technology – Advanced Diploma in ICT: Applications Development (ADICTA)

Professional Development 4 (PFD473S) – Term 3 Assessment

Project Duration

6 months (01 November 2026 – 30 April 2027)

Team Size

9 people

Location

Cape Town / Johannesburg, South Africa

Currency

South African Rand (ZAR)

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1. Project Overview / Project Charter

Project Name

Azure Cloud Migration – Retail ERP Modernisation

Project Purpose

To migrate a mid-sized South African retail company's legacy on-premise ERP system to Microsoft Azure, introduce a modern Power BI analytics layer, and implement robust security hardening that complies with POPIA.

Objectives

- Successfully migrate all core ERP modules (Finance, Inventory, Sales, Procurement, HR) to Azure with minimal business disruption.
- Achieve ≥ 99.5% system availability post go-live.
- Reduce infrastructure operating costs by at least 25% within 12 months of go-live.
- Enable real-time business intelligence via Power BI.
- Ensure full POPIA compliance and enhanced cybersecurity posture.
- Complete the project within 6 months and within the approved budget.

Scope Summary

In-scope: Assessment, planning, Azure landing zone setup, data migration, application refactoring/re-platforming, Power BI implementation, security & identity (Azure AD + Conditional Access), testing, cutover, hypercare and knowledge transfer.

Out-of-scope: Complete rewrite of custom code, mobile app development, new POS hardware, and post-project continuous improvement beyond the 30-day hypercare period.

Key Stakeholders

- Project Sponsor (CIO)
- Business Owner (Retail Operations Director)
- End users (Finance, Warehouse, Sales, HR)
- IT Operations & Security teams
- External Azure Partner (if engaged)
- Project Manager and project team

Assumptions

- Legacy system source code and documentation are available.
- Business will provide dedicated subject-matter experts (SMEs) for the duration of the project.

- Azure credits / existing Microsoft Enterprise Agreement can be leveraged.
- Load-shedding will be mitigated by dual-region Azure design and UPS at primary sites.

Constraints

- Fixed 6-month duration.
- Maximum budget of R4.85 million.
- Must comply with POPIA and internal data residency preferences.
- Limited internal cloud skills (training required).

Success Criteria

- All critical ERP functions operational on Azure with ≤ 4 hours total unplanned downtime during cutover.
- Power BI dashboards live and used by management within 2 weeks of go-live.
- Zero critical security findings at final penetration test.
- User acceptance ≥ 85%.
- Project delivered on time and within ±5% of budget.

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2. Business Justification / Project Selection

Strategic Alignment

The company is pursuing a digital transformation strategy to improve agility, reduce total cost of ownership of IT infrastructure, and enable data-driven decision-making. The current on-premise ERP is nearing end-of-life, suffers frequent performance issues, and is expensive to maintain.

Expected Benefits

- Infrastructure cost reduction of 25–35% within 12 months.
- Improved system performance and scalability for seasonal retail peaks.
- Real-time analytics enabling faster inventory and pricing decisions.
- Enhanced security and POPIA compliance posture.
- Reduced dependency on ageing hardware and specialised on-premise skills.

Costs (High-level)

Total estimated project cost: **R4.85 million** (see detailed budget in Section 7).

Alternatives Considered

1. Do nothing – rejected due to increasing risk and cost.
2. Upgrade on-premise hardware – rejected (higher long-term TCO).
3. Migrate to a different cloud provider (AWS) – rejected (existing Microsoft relationship and skills).
4. Full ERP replacement (SAP S/4HANA or Dynamics 365) – deferred to a later phase due to cost and risk.

Selected Option

Lift-and-shift + selective re-platforming to Microsoft Azure with Power BI overlay. This balances risk, cost and speed.

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3. Project Scope

In-Scope

- Discovery & assessment of current ERP landscape
- Design and implementation of Azure Landing Zone (hub-spoke)
- Migration of databases and application tiers
- Implementation of Azure AD, Conditional Access, MFA and Privileged Identity Management
- Development of 12 core Power BI dashboards
- Data migration (historical + cutover)
- Non-functional testing (performance, security, disaster recovery)
- Cutover and 30-day hypercare
- Knowledge transfer and documentation

Out-of-Scope

- Replacement of point-of-sale hardware
- Development of new mobile applications
- Full custom code rewrite
- Ongoing BAU support after hypercare

Deliverables

- Azure Landing Zone
- Migrated and hardened ERP environment
- Power BI workspace and dashboards
- Security baseline and compliance report
- Updated operational runbooks
- Project closure report

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4. Lifecycle / Delivery Approach

Selected Approach: Hybrid (Waterfall + Agile)

Justification

Cloud migration projects require strong upfront architecture and governance (Waterfall elements) while allowing iterative development of the Power BI layer, testing cycles, and configuration refinements (Agile elements).

High-level Phases

1. Initiation & Planning (Weeks 1–3) – Waterfall
2. Assessment & Design (Weeks 4–7) – Waterfall
3. Foundation Build (Landing Zone + Identity) (Weeks 8–11) – Hybrid
4. Migration Waves + Power BI Development (Weeks 12–20) – Agile sprints
5. Testing, Cutover & Hypercare (Weeks 21–26) – Waterfall with daily stand-ups

This hybrid model provides predictability for the business while retaining flexibility for technical discovery.

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5. Work Breakdown Structure (WBS)

Level 1 – Project

1. Project Management
2. Assessment & Design
3. Azure Foundation

4. Application & Data Migration
5. Analytics (Power BI)
6. Security & Compliance
7. Testing & Quality Assurance
8. Cutover & Transition
9. Project Closure

Level 2 & 3 examples (excerpt)

4. Application & Data Migration

4.1 Database migration

4.1.1 Schema assessment

4.1.2 Test migration runs

4.1.3 Production cutover scripts

4.2 Application tier migration

4.2.1 Lift-and-shift of app servers

4.2.2 Configuration & connection string updates

4.2.3 Performance tuning

(Full hierarchical WBS is maintained in the project repository and summarised in Appendix E.)

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6. Schedule, Milestones and Resource Planning

Key Milestones

Milestone	Target Date
Project Kick-off	01 Nov 2026
Azure Landing Zone Ready	15 Dec 2026

First successful test migration	20 Jan 2027
Power BI Phase 1 dashboards live	28 Feb 2027
Security baseline signed off	15 Mar 2027
Production Cutover	10 –12 Apr 2027
Hypercare End & Project Closure	30 Apr 2027

Resource Allocation (9-person team)

- Project Manager (1) – full time
- Cloud Architect (1) – full time
- Azure Engineers (2) – full time
- Data Migration Specialist (1) – full time
- Power BI Developer (1) – full time
- Security Specialist (1) – 80%
- Business Analyst / Tester (1) – full time
- Change & Training Lead (1) – 60%

Effort estimates were derived using a combination of expert judgement, analogous estimating from previous Azure migrations, and bottom-up estimating for key work packages.

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7. Cost and Budget Management

High-level Budget (ZAR)

Category	Amount (ZAR)
Internal Labour	1 850 000
External Specialist / Partner	980 000
Azure Consumption (6 months)	620 000
Software / Licences	410 000
Training & Certification	185 000
Contingency (10 %)	405 000

Total	**4 450 000**
Management Reserve	400 000
Overall Budget Ceiling	**4 850 000**

Monthly cash-flow projection is provided in Appendix B. Cost control will use earned value management (EVM) with monthly variance analysis.

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8. Strategic and Tactical Management

Governance Structure

- Project Steering Committee (Sponsor, CIO, Operations Director, PM) – monthly
- Project Board / Change Control Board – bi-weekly
- Daily stand-ups during migration waves
- Weekly technical design authority meetings

Reporting Lines

Project Manager reports directly to the CIO (Sponsor). Technical leads report to the Project Manager.

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9. Human Resource Management

Staffing Plan

All nine roles are filled by a mix of internal staff and two contracted specialists (Cloud Architect and Security Specialist).

Skills & Training

- Azure Administrator and Azure Solutions Architect training for internal engineers
- Power BI Data Analyst pathway for the BI developer
- POPIA and cloud security awareness for the whole team

Performance management will follow the organisation’s existing appraisal process with project-specific KPIs (on-time delivery of work packages, quality of deliverables, collaboration).

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10. Configuration and Change Management

Configuration Management

- All Infrastructure-as-Code (Bicep / Terraform) stored in Azure DevOps Git repositories
- Environment baselines (Dev, Test, Prod) clearly defined
- Configuration items include: Landing Zone templates, network diagrams, identity configurations, database schemas, Power BI datasets

Change Management Process

1. Change Request logged in Azure DevOps
2. Impact assessment (technical, cost, schedule, risk)
3. Review by Change Control Board
4. Approval / Rejection
5. Implementation and verification
6. Baseline update

Emergency changes follow a fast-track path with retrospective approval.

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11. Decision (Rationale) Management

Key decisions are recorded in a Decision Log with the following attributes: Decision ID, Date, Description, Alternatives considered, Rationale, Impact, Approver, Status.

Examples of major decisions already taken:

- Choice of Microsoft Azure over AWS
- Hybrid delivery approach
- Dual-region architecture for resilience against load-shedding
- Use of Azure Migrate + custom scripts rather than pure third-party tooling

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12. Risk Management

A comprehensive Risk Register is maintained (see Appendix C). Top risks include:

- Data loss or corruption during migration

- Extended downtime during cutover
- Azure cost overruns
- Skills shortages
- Security misconfigurations
- Resistance to change from business users
- Load-shedding impacting testing windows
- Vendor / partner delivery delays

Each risk has probability, impact, risk score, response strategy (Avoid / Mitigate / Transfer / Accept), contingency plan, and owner.

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13. Quality Management

Quality Objectives

- Zero Sev-1 defects in production at go-live
- System availability ≥ 99.5% in the first 30 days
- User acceptance score ≥ 85%
- All security controls aligned with Microsoft Cloud Security Benchmark and POPIA
- Complete and accurate documentation

Quality Assurance Activities

- Peer reviews of all architecture and design artefacts
- Regular quality audits against the project quality plan
- Compliance checks against ISO 9001 principles and CMM Level 2/3 practices (Requirements Management, Configuration Management, Quality Assurance, Peer Reviews)

Quality Control Activities

- Inspection of deliverables
- Formal design reviews
- Code / script reviews
- Performance, security and user acceptance testing
- Defect density and Mean Time Between Failures (MTBF) tracking

Acceptance Criteria

Defined per major deliverable and signed off by the Business Owner and Security team.

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14. Procurement Management

Major Procurements

- Microsoft Azure consumption (via existing Enterprise Agreement)
- Specialist cloud architecture and security consulting (fixed-price + time-and-materials)
- Training courses and exam vouchers
- Possible third-party migration tooling licences

Vendor selection follows the organisation's procurement policy with emphasis on Microsoft competencies and local South African presence.

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15. Project Monitoring and Reporting

- Daily stand-ups (migration phases)
- Weekly status reports to Sponsor (RAG status, progress, risks, issues, budget)
- Bi-weekly Steering Committee packs
- Monthly earned value reports
- Escalation path: PM → Sponsor → Executive Committee

Key performance metrics: Schedule Performance Index (SPI), Cost Performance Index (CPI), defect trends, and milestone achievement.

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16. Appendices

Appendix A – High-level Gantt Chart

(Described: 6-month timeline showing the five major phases, key dependencies, and milestones listed in Section 6. Critical path runs through Landing Zone → Data Migration → Cutover.)

Appendix B – Monthly Cash Flow Projection

(Detailed month-by-month breakdown of labour, Azure consumption, partner fees and

contingency drawdown – total aligning to R4.85 m ceiling.)

Appendix C – Risk Register

(16 risks with full scoring, response strategies and owners – available as a separate spreadsheet in the project repository.)

Appendix D – RACI Matrix

(9 roles × major activity groups: Initiate, Design, Build, Migrate, Test, Cutover, Close.)

Appendix E – WBS Dictionary & Resource Summary

(Effort hours per Level-2 work package and resource histogram.)

Document Control

This SPMP is a living document and will be baselined after Steering Committee approval. All subsequent changes follow the Configuration and Change Management process described in Section 10.

This completes a full, realistic, rubric-aligned Project Management Plan focused on the Cloud Migration project. You can now copy this into Word, expand any section further, or generate the detailed appendices (Gantt, Risk Register, RACI, Cash Flow) as separate files.